

Avi Kumar Singh

+44 7398 191815

avikumar.singh1508.work@gmail.com

London, UK

linkedin.com/in/avi-kumar-singh-1508/

github.com/avikumar15/

SUMMARY

Full-stack Software Engineer with 3+ years of experience building distributed systems and scalable microservices. Backend-focused in Python and Java, with a track record of delivering production-ready features end-to-end

TECHNICAL SKILLS

Languages/Frameworks (Advanced): Java, Python, JavaScript, Flask, Spring Boot, ReactJS, MySQL

Languages/Frameworks (Intermediate): C++, Android, DynamoDB, PostgreSQL

Core Competency: Distributed Systems, REST API Design, Microservices, System design, Cloud Native development

Tools: Git, Docker, Unix, AWS, Azure, Jenkins, Agile

Certification: AWS Technical Essentials by AWS, Introduction to DevOps Tools by Simplilearn

EXPERIENCE

V-Nova

September 2024 – Present

Software Engineer

London, UK

- Engineered a distributed orchestration framework in Python to parallelise compute-intensive workloads across distributed worker nodes.
- Optimised webpage UI performance by reducing latency by ~99% via on-demand loading and Redis caching
- Scaled horizontally from 10 nodes to ~150 AWS nodes.
- Built an LLM-powered framework for encoder and log analysis, eliminating manual effort.
- Skills:** Python, Java, JavaScript, Flask, Docker, Ansible, AWS, Redis, MySQL

Espire Infolabs

May 2023 – September 2024

Software Engineer

London, UK

- Engineered dynamic document engine using Spring Boot and React JS, reducing licensing costs.
- Built a Java app on Azure Functions to automate CCM framework migration, cutting manual effort by 70%
- Skills:** Java, Spring boot, React JS, MySQL, Docker, Azure, Machine Learning

Amazon

August 2022 – February 2023

Software Engineer

Cambridge, UK

- Maintained and enhanced a high-availability Java-based microservice powering Alexa Linked Data queries, improving reliability and onboarding new use cases with comprehensive unit testing.
- Migrated dependency service using shadow mode approach resulting in reduction of fault rates.
- Skills:** Java, Jersey, AWS, React JS, DynamoDB, Git, TDD

EDUCATION

National Institute of Technology Trichy

July 2018 – July 2022

Bachelor of Technology in Computer Science, Minor in Economics

INTERNSHIP

Oracle

June 2021 – July 2021

Intern – Application Engineer

Bengaluru, India

- Developed Capacity Manager, a cost-effective utility tool using the Spring framework and leveraging Oracle Cloud saving upto 5% pod provisioning failures. Developed a ReactJS web app for real time cloud insights.
- Crafted a Java cron job script integrated with slackbot to automate cloud related alerts to admin
- Skills:** Java, Springboot, React JS, Git, OCI, Postgresql

DailyObjects

December 2019 – January 2020

Intern – Android Engineer

New Delhi, India

- Developed new UI views of the Android app using Kotlin and Jetpack libraries, adopting a TDD approach, impacting over 500,000 app users. Automated testing by leveraging Firebase Test Lab and CircleCI.

PROJECTS

Quick Mail CLI | Python, Docker, PyPI | GitHub | PyPI

Spring 2021

- Developed and dockerised command line interface to send email quickly. Averages 35 installs every month.

Stock Price Prediction | Python, NLP, Statistical Modelling – ARIMA, Sentiment Analysis

Spring 2020

- Trained an ARIMA-based stock price forecasting model and also a model based on sentiment analysis
- Performed sentiment analysis by scrapping stock news from FinViz using Python's BeautifulSoup.